



Ascham School

Mathematics Extension 1

Trial HSC Examination

Monday 25th July 2016

2 hours

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using black pen.
- Board-approved calculators may be used.
- A Reference Sheet is provided.
- In Questions 11–14, show relevant mathematical reasoning and/or calculations.

Total marks – 70

Section I Pages 2–4

10 marks

- Use the Multiple Choice Answer Sheet provided to answer Q1-10.
- Allow about 15 minutes for this section.

Section II Pages 5–9

60 marks

- Answer Questions 11-14.
- Answer each question in a new booklet.
- Label all sections clearly with your name/number and teacher's initials.
- Allow about 1 hour and 45 minutes for this section.

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 If α , β and γ are roots of the equation $x^3 - 3x^2 + 4x + 2 = 0$, find the value of $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$.

(A) 4 (B) $\frac{1}{4}$ (C) $-\frac{2}{3}$ (D) $-\frac{3}{2}$

- 2 O is the centre of the circle below. Which is the correct value for θ ?

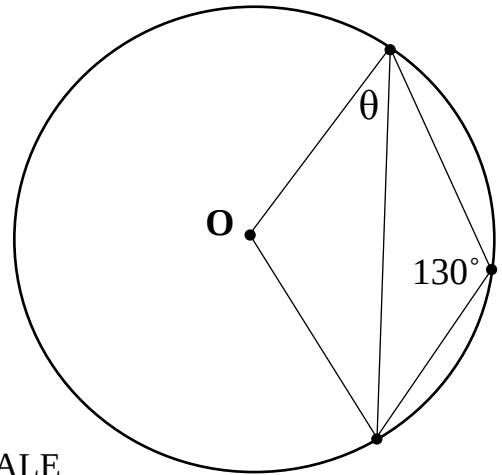
(A) 25°

(B) 40°

(C) 50°

(D) 65°

NOT TO SCALE



- 3 What are the asymptotes of $y = \frac{1 - x^2}{x^2 - 4}$?

(A) $x = 0$, $x = 2$, $y = -1$

(B) $x = 2$, $x = -2$, $y = 1$

(C) $x = 2$, $x = -2$, $y = -1$

(D) $x = 2$, $x = -2$, $y = 0$

4 The equation of the normal to the curve $x^2 = 20y$ at the point $(10p, 5p^2)$ is:

(A) $x + py = 5p^3 + 10p$

(B) $x - py = 5p^3 - 10p$

(C) $px + y = 15p^2$

(D) $px - y + 15p^2 = 0$

5 Choose the correct value of $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{3x}$.

(A) $\frac{1}{6}$

(B) $\frac{1}{3}$

(C) $\frac{1}{2}$

(D) 6

6 The angle θ satisfies $\sin \theta = \frac{1}{\sqrt{5}}$ and $\frac{\pi}{2} < \theta < \pi$.

What is the exact value of $\sin 2\theta$?

(A) $\frac{4}{5}$

(B) $\frac{4}{\sqrt{5}}$

(C) $-\frac{4}{5}$

(D) $-\frac{4}{\sqrt{5}}$

7 A particle has displacement function $x = 3\cos 5t$. Its acceleration can be written as:

(A) $\ddot{x} = 9x$

(B) $\ddot{x} = -9x$

(C) $\ddot{x} = 25x$

(D) $\ddot{x} = -25x$

8 The velocity, v metres per second, of a particle moving in simple harmonic motion along the x -axis is given by the equation $v^2 = 100 - 16x^2$.

What is the amplitude, in metres, of the motion of the particle?

(A) $\frac{2}{5}$

(B) $2\frac{1}{2}$

(C) 4

(D) 10

- 9 Select the equation which could represent the graph below:

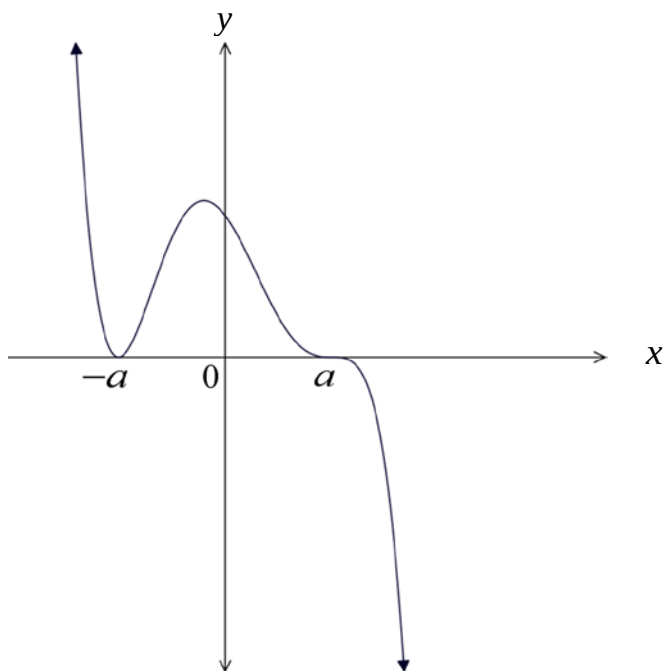
DIAGRAM NOT TO SCALE

(A) $y = (x + a)^2(a - x)^3$

(B) $y = (x - a)^2(x + a)^3$

(C) $y = (x + a)^2(x - a)^3$

(D) $y = (x - a)^2(a - x)^3$



- 10 Which of the following represents the derivative of $\cos^{-1}\left(\frac{1}{x}\right)$?

(A) $\frac{1}{\sqrt{x^2 - 1}}$

(B) $\frac{-1}{\sqrt{x^2 - 1}}$

(C) $\frac{1}{x\sqrt{x^2 - 1}}$

(D) $\frac{-1}{x\sqrt{x^2 - 1}}$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

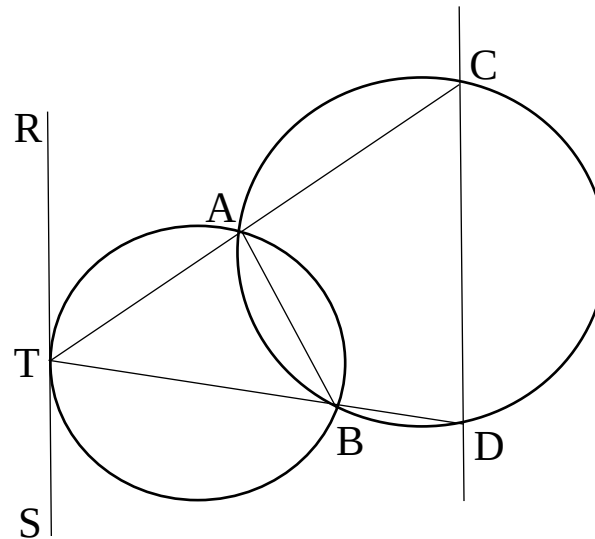
Answer each question in a SEPARATE writing booklet. Extra writing booklets are available. In Questions 11–14, you should include relevant mathematical reasoning and/ or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the value of k if $x^3 + 3x^2 - 4x + k$ is divisible by $x + 2$. 2
- (b) Solve the inequality $\frac{x}{x-5} \geq 2$. 3
- (c) The lines $y = mx + 5$ and $y = -3x + 7$ are inclined to each other at an angle of 45° .
- (i) Show that $\left| \frac{m+3}{1-3m} \right| = 1$. 1
- (ii) Hence find the possible values of m . 2
- (d) Find the general solutions to the equation: $\sin 2\theta = \sin \theta$. 3
- (e) Consider the graph of the function $f(x) = x^2 + 2x$ for $x \leq -1$.
- (i) Find the equation of $y = f^{-1}(x)$ and state its domain. 2
- (ii) Sketch the graphs of $y = f(x)$ and its inverse function $y = f^{-1}(x)$ on the same number plane. 2

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) In the diagram, two unequal circles intersect at A and B. The line RS is tangential to the smaller circle at T. The lines TA and TB meet the larger circle at C and D respectively.



NOT TO SCALE

Copy the diagram into your exam booklet.

- (i) State a theorem to explain why $\angle BAT = \angle BDC$. 1
- (ii) Prove that $RS \parallel CD$. 3
- (b) Consider the equation $x^2 - 9 + \log_e x = 0$.
- (i) By drawing the graph of $y = \log_e x$ and another appropriate graph on the same axes, explain why the equation has only one root. 1
- (ii) Show, using calculations, that the root of the equation lies between 2 and 3. 2
- (iii) Taking $x_0 = 2.5$ as the first approximation, use Newton's method to find a second approximation of the root correct to 2 decimal places. 2
- (c) (i) Use the substitution $x = \sin \theta$ to show that $\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx = \int_0^{\frac{\pi}{6}} \sin^2 \theta d\theta$. 3
- (ii) Hence find the exact value of $\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx$. 3

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Show that $\sqrt{3}\cos\theta + \sin\theta = 2\cos\left(\theta - \frac{\pi}{6}\right)$. 2

(ii) Hence solve the equation $\sqrt{3}\cos\theta + \sin\theta = -1$ for $0 \leq \theta \leq 2\pi$. 3

A particle moves in simple harmonic motion along a straight line so that its displacement, x metres, at time t seconds, is given by:

$$x = \sqrt{3}\cos\left(\frac{t}{3}\right) + \sin\left(\frac{t}{3}\right).$$

(iii) Find the smallest positive value of t for which $x = -1$. 1

(iv) Find the distance travelled by the particle in going from its initial position to the position $x = -1$. Justify your answer. 3

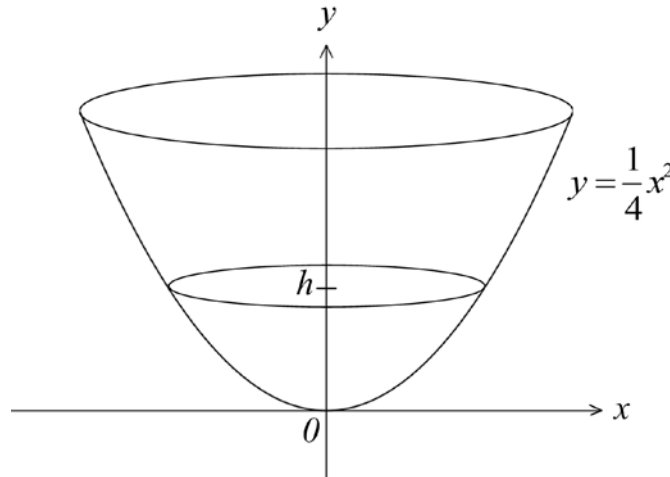
(b) The velocity of a particle is given by the equation $v = \frac{1}{e^x}$.

If the initial displacement is $x = 0$, find the equation for the displacement x , in terms of t . 3

(c) Prove by mathematical induction that $\sum_{r=1}^n \log\left(\frac{r+1}{r}\right) = \log(n+1)$. 3

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) A large industrial container is in the shape of a paraboloid, which is formed by rotating part of the parabola $y = \frac{1}{4}x^2$ around the y -axis, as shown in the diagram.
Liquid is poured into the container at the rate of 2 m^3 per minute.



- (i) Show that the volume $V \text{ m}^3$ of liquid in the container when the depth of liquid is h metres, is given by $V = 2\pi h^2$. 1
- (ii) At what rate is the height (h) of the liquid rising when the depth is 0.5 metres? 3
- (iii) If the container is 3 metres high, how long will it take to fill the container? 1
- (b) $P(2t, t^2)$ is a point on the parabola $x^2 = 4y$ with focus $S(0, 1)$.
The point M divides the interval SP externally in the ratio $3 : 1$.
- (i) Show that the coordinates of the point M are $\left(3t, \frac{3t^2 - 1}{2}\right)$. 2
- (ii) Hence show that as P moves on the parabola $x^2 = 4y$, then M moves on the parabola $x^2 = 6y + 3$. 2

Question 14 continues on page 9

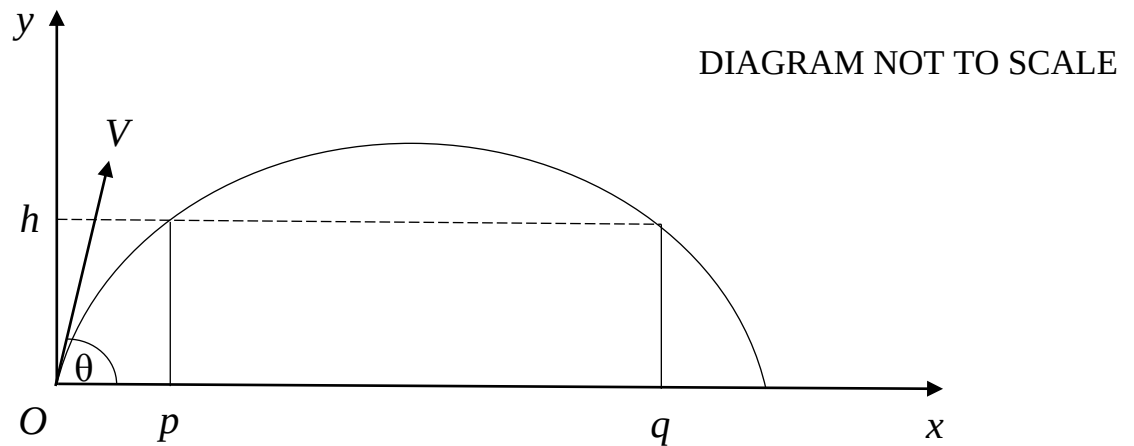
- (c) A particle is projected with initial velocity $V \text{ ms}^{-1}$ at an angle of elevation θ from the origin O . The particle just clears two vertical chimneys of height h metres at horizontal distances p metres and q metres from O .

The diagram shows the path of this projectile where the acceleration due to gravity is taken as 10 ms^{-2} and air resistance is ignored.

You may assume that the projectile's trajectory is defined by the equations:

$$x = Vt \cos \theta \quad \text{and} \quad y = Vt \sin \theta - 5t^2 \quad [\text{Do not prove these.}]$$

where x and y are the horizontal and vertical displacements of the projectile in metres at time t seconds.



- (i) Show that $V^2 = \frac{5p^2(1 + \tan^2 \theta)}{p \tan \theta - h}$. 3
- (ii) Hence show that $\tan \theta = \frac{h(p + q)}{pq}$. 3

END OF PAPER

SECTION I

$$\textcircled{1} \quad \frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$$

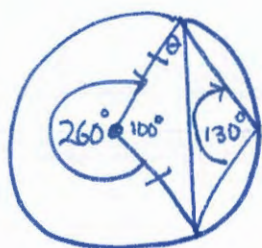
$$= \frac{\gamma + \alpha + \beta}{\alpha\beta\gamma}$$

$$= \frac{-b/a}{-d/a} \quad \text{where } \begin{array}{l} a=1 \\ b=-3 \\ c=4 \\ d=2 \end{array}$$

$$= \frac{-3/1}{-2/1}$$

$$= -\frac{3}{2} \quad \therefore \underline{D}$$

②



$$\theta = \frac{180^\circ - 100^\circ}{2}$$

$$\theta = 40^\circ$$

$$\therefore \underline{B}$$

$$\textcircled{3} \quad y = \frac{1-x^2}{(x-2)(x+2)}$$

$$x-2 \neq 0$$

$$x+2 \neq 0$$

$\therefore$ Vertical Asymptotes at $x = \pm 2$

$$\lim_{x \rightarrow \infty} \frac{1-x^2}{x^2-4} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2} - \frac{x^2}{x^2}}{\frac{x^2}{x^2} - \frac{4}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2} - 1}{1 - \frac{4}{x^2}}$$

$$= \frac{0-1}{1-0}$$

$$= -1$$

$\therefore$ Horizontal Asymptote at $y = -1$

$$\therefore \underline{C}$$

$$\textcircled{4} \quad x^2 = 20y \quad \text{or} \quad y = \frac{x^2}{20}$$

$$\frac{dy}{dx} = \frac{x}{10}$$

$$\text{At } x = 10p: \quad \frac{dy}{dx} = \frac{10p}{10} = p$$

$$\therefore \text{Gradient of normal} = -\frac{1}{p}$$

Equation of normal is:

$$y - 5p^2 = -\frac{1}{p}(x - 10p)$$

$$py - 5p^3 = -x + 10p$$

$$\therefore x + py = 5p^3 + 10p$$

$$\therefore \underline{A}$$

$$\textcircled{5} \quad \lim_{x \rightarrow 0} \frac{\sin^{x/2}}{3x}$$

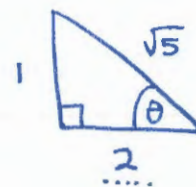
$$= \lim_{x \rightarrow 0} \frac{\sin^{x/2}}{x/2} \times \frac{x/2}{3x}$$

$$= 1 \times \frac{1}{6}$$

$$= \frac{1}{6}$$

$$\therefore \underline{A}$$

$$\textcircled{6} \quad \text{If } \sin \theta = \frac{1}{\sqrt{5}} \quad \text{and } \pi/2 < \theta < \pi$$



$$\cos \theta = -\frac{2}{\sqrt{5}}$$

$$\text{so } \sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2 \times \frac{1}{\sqrt{5}} \times -\frac{2}{\sqrt{5}}$$

$$= -\frac{4}{5}$$

$$\therefore \underline{C}$$

$$\textcircled{7} \quad x = 3 \cos 5t$$

$$\dot{x} = -15 \sin 5t$$

$$\ddot{x} = -75 \cos 5t$$

$$\ddot{x} = -25(3 \cos 5t)$$

$$\ddot{x} = -25x$$

$$\therefore \underline{D}$$

⑧ When $v=0$:
 $100 - 16x^2 = 0$
 $(10 - 4x)(10 + 4x) = 0$
 $\therefore x = \pm \frac{5}{2}$ are endpoints
 $\therefore$ centre of motion is $x=0$
 and amplitude $= \frac{5}{2} \therefore \underline{B}$

⑨ $y = -(x+a)^2(x-a)^3$
 $= (x+a)^2(a-x)^3 \therefore \underline{A}$

⑩ $\frac{d}{dx} [\cos^{-1}(x^{-1})]$
 $= \frac{-1}{\sqrt{1 - (x^{-1})^2}} \times -x^{-2}$
 $= \frac{-1}{\sqrt{1 - \frac{1}{x^2}}} \times \frac{-1}{x^2}$
 $= \frac{1}{x^2 \sqrt{\frac{x^2 - 1}{x^2}}}$
 $= \frac{1}{\frac{x^2}{\sqrt{x^2}} \sqrt{x^2 - 1}}$
 $= \frac{1}{x \sqrt{x^2 - 1}} \therefore \underline{C}$

SECTION II

Question 11

a) If $P(x) = x^3 + 3x^2 - 4x + k$
 then $P(-2) = 0$
 $(-2)^3 + 3(-2)^2 - 4(-2) + k = 0$
 $-8 + 12 + 8 + k = 0$
 $\therefore k = -12$

b) $\frac{x}{x-5} \geq 2 \quad [x \neq 5]$

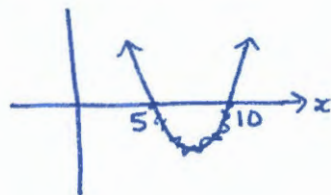
$$\frac{x(x-5)^2}{x-5} \geq 2(x-5)^2$$

$$x(x-5) \geq 2(x-5)^2$$

$$2(x-5)^2 - x(x-5) \leq 0$$

$$(x-5)[2(x-5) - x] \leq 0$$

$$(x-5)(x-10) \leq 0$$



$$\therefore \underline{5 < x \leq 10}$$

c) i) $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$$\tan 45^\circ = \left| \frac{m - 3}{1 + m(-3)} \right|$$

$$1 = \left| \frac{m+3}{1-3m} \right|$$

$$\therefore \left| \frac{m+3}{1-3m} \right| = 1$$

(as required)

$$(i) \frac{m+3}{1-3m} = 1 \quad \text{or} \quad \frac{m+3}{1-3m} = -1$$

$$m+3 = 1-3m$$

$$4m = -2$$

$$m = -\frac{1}{2}$$

$$m+3 = -(1-3m)$$

$$m+3 = -1+3m$$

$$4 = 2m$$

$$m = 2$$

$$\therefore \underline{m = -\frac{1}{2} \quad \text{or} \quad m = 2}$$

$$d) \sin 2\theta = \sin \theta$$

$$2\sin \theta \cos \theta - \sin \theta = 0$$

$$\sin \theta (2\cos \theta - 1) = 0$$

$$\sin \theta = 0 \quad \text{or} \quad \cos \theta = \frac{1}{2}$$



$$\therefore \theta = \pi n \quad \text{or} \quad \pm \frac{\pi}{3} + 2n\pi$$

$$(n \in \mathbb{Z})$$

$$e) i) \text{ Let } y = f(x)$$

$$\text{i.e. } y = x^2 + 2x \quad \text{where } x \leq -1$$

$$\therefore \text{Inverse: } x = y^2 + 2y$$

Completing the square...

$$x+1 = y^2 + 2y + 1$$

$$x+1 = (y+1)^2$$

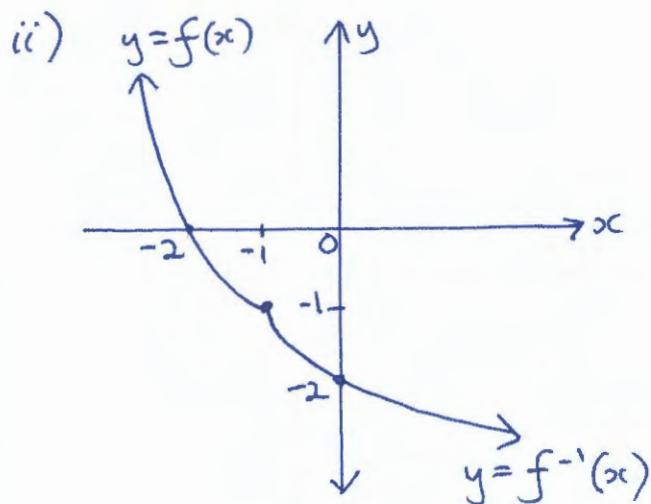
$$\pm \sqrt{x+1} = y+1$$

$$y = -1 \pm \sqrt{x+1}$$

$$\text{but } y \leq -1$$

$$\therefore f^{-1}(x) = -1 - \sqrt{x+1}$$

over the domain $x \geq -1$



Question 12

a) i) The exterior $\angle$ of a cyclic quadrilateral equals the interior opposite $\angle$.

ii) $\angle BAT = \angle BTS$
($\angle$ between tangent and chord)
(equals $\angle$ in alternate segment)

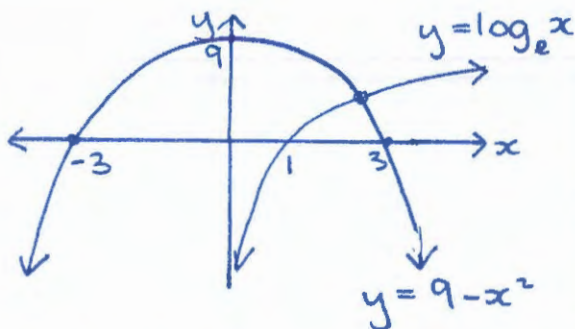
$$\therefore \angle BTS = \angle BDC$$

$\therefore RS \parallel CD$
(alternate $\angle$ s equal)

b) i) $x^2 - 9 + \log_e x = 0$

$$\text{or } \log_e x = 9 - x^2$$

The solution represented by the x -value of the point of intersection of the graphs $y = \log_e x$ and $y = 9 - x^2$



As there is only one point of intersection, there is only one root of the equation $x^2 - 9 + \log_e x = 0$

(i) Let $f(x) = x^2 - 9 + \log_e x$

$$f(2) = 2^2 - 9 + \log_e 2 = -4.306... < 0$$

$$f(3) = 3^2 - 9 + \log_e 3 = 1.098... > 0$$

Since $f(2) < 0$ and $f(3) > 0$, the root must lie between 2 and 3, given $f(x)$ is continuous.

(ii) If $f(x) = x^2 - 9 + \log_e x$

$$\text{then } f'(x) = 2x + \frac{1}{x}$$

$$\text{Using: } x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$\text{where } x_0 = 2.5,$$

$$x_1 = 2.5 - \frac{2.5^2 - 9 + \ln 2.5}{2(2.5) + \frac{1}{2.5}}$$

$$= 2.839...$$

$$\therefore x_1 \approx 2.84 \text{ (2 decimal places)}$$

c) i) If $x = \sin \theta$ then $\frac{dx}{d\theta} = \cos \theta$

$$\text{i.e. } dx = \cos \theta d\theta$$

$$\text{and } \int \frac{x^2}{\sqrt{1-x^2}} dx = \int \frac{\sin^2 \theta}{\sqrt{1-\sin^2 \theta}} \cos \theta d\theta$$

$$= \int \frac{\sin^2 \theta}{\sqrt{\cos^2 \theta}} \cos \theta d\theta$$

$$= \int \sin^2 \theta d\theta$$

$$\text{Also when } x=0: \sin \theta = 0$$

$$\therefore \theta = 0$$

$$\text{and when } x = \frac{1}{2}: \sin \theta = \frac{1}{2}$$

$$\therefore \theta = \pi/6$$

$$\therefore \int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx = \int_0^{\pi/6} \sin^2 \theta d\theta$$

(ii) $\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx$

$$= \int_0^{\pi/6} \sin^2 \theta d\theta = \frac{1}{2} \int_0^{\pi/6} 1 - \cos 2\theta d\theta$$

$$= \frac{1}{2} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi/6}$$

$$= \frac{1}{2} \left[\frac{\pi}{6} - \frac{\sin(2 \times \frac{\pi}{6})}{2} - \left(0 - \frac{\sin(2 \times 0)}{2} \right) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{6} - \frac{\sqrt{3}/2}{2} - 0 \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right]$$

$$= \frac{1}{2} \times \frac{1}{2} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right]$$

$$= \frac{1}{4} \left(\frac{2\pi - 3\sqrt{3}}{6} \right) \text{ or } \underline{\underline{\frac{2\pi - 3\sqrt{3}}{24}}}$$

Question 13

a) i) show that:

$$\sqrt{3} \cos \theta + \sin \theta = 2 \cos \left(\theta - \frac{\pi}{6} \right)$$

$$\text{RHS} = 2 \cos \left(\theta - \frac{\pi}{6} \right)$$

$$= 2 \left(\cos \theta \cos \frac{\pi}{6} + \sin \theta \sin \frac{\pi}{6} \right)$$

$$= 2 \cos \theta \left(\frac{\sqrt{3}}{2} \right) + 2 \sin \theta \left(\frac{1}{2} \right)$$

$$= \sqrt{3} \cos \theta + \sin \theta$$

$$= \text{LHS}$$

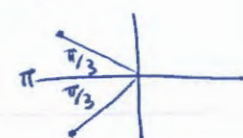
ii) Solving $\sqrt{3} \cos \theta + \sin \theta = -1$

is equivalent to solving

$$2 \cos \left(\theta - \frac{\pi}{6} \right) = -1$$

$$\cos \left(\theta - \frac{\pi}{6} \right) = -\frac{1}{2}$$

related $\angle = \frac{\pi}{3}$



$$\theta - \frac{\pi}{6} = 2\frac{\pi}{3} \quad \text{or} \quad \theta - \frac{\pi}{6} = 4\frac{\pi}{3}$$

$$\theta = 2\frac{\pi}{3} + \frac{\pi}{6} \quad \theta = 4\frac{\pi}{3} + \frac{\pi}{6}$$

$$\therefore \theta = 5\frac{\pi}{6} \quad \text{or} \quad \theta = 3\frac{\pi}{2}$$

iii) From part (ii), the smallest positive solution to

$$\sqrt{3} \cos \theta + \sin \theta = -1 \text{ is } \theta = 5\frac{\pi}{6}$$

$\therefore$ The smallest positive solution to

$$\sqrt{3} \cos \left(\frac{t}{3} \right) + \sin \left(\frac{t}{3} \right) = -1 \text{ is when}$$

$$\frac{t}{3} = 5\frac{\pi}{6}$$

$$t = 5\frac{\pi}{6} \times 3$$

$$\therefore \underline{t = 5\frac{\pi}{2} \text{ seconds}}$$

iv) Using $x = 2 \cos \left(\frac{t}{3} - \frac{\pi}{6} \right)$

When $t=0$: $x = 2 \cos \left(\frac{0}{3} - \frac{\pi}{6} \right)$

$$= 2 \cos \left(-\frac{\pi}{6} \right)$$

$$= 2 \cos \frac{\pi}{6}$$

$$= 2 \frac{\sqrt{3}}{2}$$

$$\therefore \text{initial position} = \sqrt{3}$$

Now we see that the

amplitude = 2 $\therefore$ particle

oscillates between $x = -2$ and $+2$

However we need to find the initial direction of motion

i.e. Find velocity when $t=0$

$$\dot{x} = -\frac{1}{3} \times 2 \sin \left(\frac{t}{3} - \frac{\pi}{6} \right)$$

and when $t=0$:

$$\dot{x} = -\frac{2}{3} \sin \left(0 - \frac{\pi}{6} \right)$$

$$= +\frac{2}{3} \sin \left(\frac{\pi}{6} \right)$$

$$> 0$$

$\therefore$ Particle initially moves to the right



$\therefore$ Distance travelled

$$= (2 - \sqrt{3}) + 3$$

$$= \underline{5 - \sqrt{3} \text{ metres}}$$

$$b) \quad v = \frac{1}{e^x}$$

$$\text{i.e.} \quad \frac{dx}{dt} = \frac{1}{e^x}$$

Taking reciprocals of both sides:

$$\frac{dt}{dx} = e^x$$

Taking integrals of both sides:

$$\int \frac{dt}{dx} = \int e^x dx$$

$$t = e^x + c$$

When $t=0$, $x=0$

$$\therefore 0 = e^0 + c$$

$$0 = 1 + c$$

$$\therefore c = -1$$

$$\text{So } t = e^x - 1$$

$$e^x = t + 1$$

$$\ln e^x = \ln(t+1)$$

$$\therefore \underline{x = \ln(t+1)}$$

$$c) \text{ Prove } \sum_{r=1}^n \log\left(\frac{r+1}{r}\right) = \log(n+1)$$

by Mathematical Induction

Prove true for $n=1$:

$$\begin{aligned} \text{LHS} &= \log\left(\frac{1+1}{1}\right) & \text{RHS} &= \log(1+1) \\ &= \log 2 & &= \log 2 \end{aligned}$$

$\therefore$ since LHS = RHS, it is true for $n=1$.

Assume true for $n=k$:

$$\text{i.e.} \quad \sum_{r=1}^k \log\left(\frac{r+1}{r}\right) = \log(k+1)$$

$$\begin{aligned} \text{i.e.} \quad & \log\left(\frac{2}{1}\right) + \log\left(\frac{3}{2}\right) + \log\left(\frac{4}{3}\right) + \dots + \log\left(\frac{k+1}{k}\right) \\ &= \log(k+1) \quad \quad \quad * \end{aligned}$$

Prove true for $n=k+1$:

$$\text{ATP:} \quad \sum_{r=1}^{k+1} \log\left(\frac{r+1}{r}\right) = \log(k+2)$$

$$\begin{aligned} \text{LHS} &= \underbrace{\log\left(\frac{2}{1}\right) + \log\left(\frac{3}{2}\right) + \dots + \log\left(\frac{k+1}{k}\right)}_{\log(k+1)} + \log\left(\frac{k+2}{k+1}\right) \\ &= \log(k+1) + \log\left(\frac{k+2}{k+1}\right) \quad \text{using } * \text{ assumption} \\ &= \log\left[(k+1)\left(\frac{k+2}{k+1}\right)\right] \\ &= \log(k+2) \\ &= \text{RHS} \end{aligned}$$

$\therefore$ The proposition is proved true by Mathematical Induction.

Question 14

a) i) $y = \frac{1}{4}x^2 \Rightarrow 4y = x^2$

$$V = \pi \int_0^h 4y \, dy$$

$$= \pi \left[\frac{4y^2}{2} \right]_0^h$$

$$= \pi [2h^2 - 2(0)^2]$$

$\therefore V = 2\pi h^2$ (as required)

ii) $\frac{dV}{dh} = 4\pi h$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$\frac{dh}{dt} = \frac{1}{4\pi h} \times 2$$

$$\therefore \frac{dh}{dt} = \frac{1}{2\pi h}$$

when $h = 0.5\text{m}$,

$$\frac{dh}{dt} = \frac{1}{2\pi(0.5)}$$

$$= \frac{1}{\pi} \text{ m/min}$$

iii) Using $V = 2\pi h^2$

when $h = 3$:

$$V = 2\pi(3)^2$$

$$V = 18\pi \text{ m}^2$$

Now given the fill rate is 2m^3 per minute the time to fill the container

$$= 18\pi \div 2$$

$$= \underline{9\pi \text{ minutes}}$$

b) i) Using

$$x = \frac{lx_1 + kx_2}{k+l}$$

$$\text{and } y = \frac{ly_1 + ky_2}{k+l}$$

where $k:l = 3:-1$

and $(x_1, y_1) = S(0, 1)$

and $(x_2, y_2) = P(2t, t^2)$

$$x = \frac{-1(0) + 3(2t)}{3+(-1)} \quad y = \frac{-1(1) + 3(t^2)}{3+(-1)}$$

$$x = \frac{6t}{2} = 3t \quad \text{and } y = \frac{3t^2-1}{2}$$

$\therefore M$ is $\left(3t, \frac{3t^2-1}{2}\right)$ as required.

ii) From the coordinates of M ,

$$x = 3t \quad \text{i.e.} \quad t = \frac{x}{3} \dots (1)$$

$$y = \frac{3t^2-1}{2} \dots (2)$$

sub. (1) into (2):

$$y = \frac{3\left(\frac{x}{3}\right)^2 - 1}{2}$$

$$2y = 3\left(\frac{x^2}{9}\right) - 1$$

$$2y+1 = \frac{x^2}{3}$$

$$3(2y+1) = x^2$$

$\therefore x^2 = 6y+3$ as required.

c) i) Given $x = Vt \cos \theta \dots (1)$
and $y = Vt \sin \theta - 5t^2 \dots (2)$

If the point (p, h) satisfies the equations, then

substituting $x = p$ into (1):

$$p = Vt \cos \theta$$

i.e. $t = \frac{p}{V \cos \theta}$

and substituting $y = h$ into (2):

$$h = Vt \sin \theta - 5t^2$$

$$\therefore h = V \left(\frac{p}{V \cos \theta} \right) \sin \theta - 5 \left(\frac{p}{V \cos \theta} \right)^2$$

$$h = p \tan \theta - \frac{5p^2}{V^2} \sec^2 \theta$$

$$h = p \tan \theta - \frac{5p^2}{V^2} (1 + \tan^2 \theta)$$

$$\frac{5p^2}{V^2} (1 + \tan^2 \theta) = p \tan \theta - h$$

$$5p^2 (1 + \tan^2 \theta) = V^2 (p \tan \theta - h)$$

$$\therefore V^2 = \frac{5p^2 (1 + \tan^2 \theta)}{p \tan \theta - h}$$

(as required)

ii) since the point (q, h) satisfies the equations too,

$$\text{then } V^2 = \frac{5q^2 (1 + \tan^2 \theta)}{q \tan \theta - h}$$

Equating expressions for V^2 :

$$\frac{5p^2 (1 + \tan^2 \theta)}{p \tan \theta - h} = \frac{5q^2 (1 + \tan^2 \theta)}{q \tan \theta - h}$$

$$\frac{p^2}{p \tan \theta - h} = \frac{q^2}{q \tan \theta - h}$$

$$p^2 (q \tan \theta - h) = q^2 (p \tan \theta - h)$$

$$p^2 q \tan \theta - p^2 h = p q^2 \tan \theta - q^2 h$$

$$p^2 q \tan \theta - p q^2 \tan \theta = p^2 h - q^2 h$$

$$p q \tan \theta (p - q) = h (p - q) (p + q)$$

$$p q \tan \theta = h (p + q)$$

$$\therefore \tan \theta = \frac{h (p + q)}{p q}$$

(as required)